

Mohammad Hussein Ghafoori



// Backend & AI Engineer

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<https://mohammad-hussein-dev.github.io/mohammad-hussein-dev/>

> PROFESSIONAL SUMMARY

Backend & AI Engineer with a Physics & Mathematics foundation. I architect scalable systems with Django, FastAPI, and PostgreSQL — reducing latency by 40% and handling 1.2K concurrent requests. Physics-inspired problem solver delivering clean, testable code.

> EXPERIENCE

Backend Developer & AI Engineer @ Self Employed

2023-06 –

- Designed and deployed 5 production-grade RESTful APIs with Django and DRF, handling 1.2K concurrent requests and reducing latency by 40% through query optimization and Redis caching.
- Engineered a Genetic Algorithm pipeline processing 5,000+ spatial data points, achieving 98% convergence accuracy under multi-objective constraints.
- Architected secure backend systems with JWT authentication, RBAC, and bilingual (EN/FA) interfaces with 98% test coverage and CI/CD via GitHub Actions.

Tech Stack: Python | Django | DRF | FastAPI | PostgreSQL | Redis | Docker | Git | GitHub Actions | Linux

\$ Reduced API latency by 40% & handled 1.2K requests | 98% GA convergence | 12-min deployments via CI/CD

> OPEN SOURCE CONTRIBUTIONS

Open Source Contributions

- django-boilerplate : Production-ready Dockerization (multi-stage builds, PostgreSQL). Reduced setup time from 30 min to 2 min.
- studybuild-customer-analysis-02 : Customer Behavior Analysis with RFM, Pareto (80/20), and churn risk detection.
- linear-regression-project : Major documentation overhaul, CI setup, and code quality improvements (PEP 8, docstrings).

> PROJECTS

Electro-Thermal PINN

2026-07 – Present

Code : <https://github.com/mohammad-hussein-dev/electro-thermal-pinn>

Electro-Thermal PINN: PINN solving Maxwell's equations & heat equation. 3 architectures (MLP, MLPPINN, TransformerPINN). L2 error < 9e-04. PyTorch, Python, NumPy, Matplotlib.

Tech Stack: [PyTorch](#) | [Python](#) | [NumPy](#) | [Matplotlib](#) | [Transformers \(RoPE, GQA, SwiGLU\)](#)

Django Task Manager

2026-07 – Present

Code : <https://github.com/mohammad-hussein-dev/django-task-manager>

Django Task Manager: Production-ready task mgmt with JWT, RBAC, bilingual UI. 96% test coverage, sub-200ms response, Docker + CI/CD. Django, DRF, PostgreSQL, Redis, Celery, pytest.

Tech Stack: [Django](#) | [DRF](#) | [PostgreSQL](#) | [Redis](#) | [Celery](#) | [Docker](#) | [pytest](#) | [GitHub Actions](#)

Projectile Physics Simulator

2026-06 – Present

Code : <https://github.com/mohammad-hussein-dev/projectile-physics-simulator>

Projectile Physics Simulator: 2D projectile simulator with 100% test coverage, bilingual output, NumPy vectorization. Python, NumPy, Matplotlib, pytest.

Tech Stack: [Python](#) | [NumPy](#) | [Matplotlib](#) | [pytest](#)

> SKILLS

Programming Languages: Python | JavaScript

Frameworks & Libraries: Django | Django REST Framework | FastAPI

Databases & Cache: PostgreSQL | Redis | SQLite

DevOps & Tools: Docker | Git | GitHub Actions | Linux (Arch)

Scientific & ML: PyTorch | NumPy | Matplotlib | Pandas | SymPy

> LANGUAGES

Persian: Native

English: Professional Working